Mehul Mangave

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EDUCATION

San Jose State University — **San Jose, CA, USA**
Master of Science in Computer Engineering — May 2026
Vishwakarma Institute of Technology — **Pune, India**
Bachelor of Technology in Electronics and Telecommunications Engineering — May 2023
Relevant coursework: Designing Data Structures, Algorithm Design, Operating Systems, Advanced Computer Design, Systems Software, Artificial Intelligence and Data Engineering.

SKILLS

Languages:	Python, C/C++, Java, Bash, SQL, Go
Cloud and containers:	AWS (EC2, S3, RDS, DynamoDB, Lambda, ECS/ECR, API Gateway, ELB), GCP, Azure, Kubernetes, Docker
IaC & Policy:	Terraform, Terraform Cloud, Sentinel (policy-as-code), CloudWatch, Prometheus, Grafana
Applied AI and LLM:	RAG pipelines, vector embeddings, knowledge graphs, AWS Bedrock, Vertex AI, PyTorch, TensorFlow

WORK EXPERIENCE

State Street Bank and Trust, Boston, US — June 2025 – Aug 2025
Cloud & AI Infrastructure Engineer Intern

- Built a production RAG system over internal Confluence documentation with Python backend APIs, giving engineers self-serve answers and cutting repeat support questions to the platform team.
- Engineered a **Terraform automation** proof of concept (PoC) that dynamically generates infrastructure templates based on user-defined requirements, utilizing in-house modules to streamline deployment workflows and reduce manual effort by over **60**%.
- Collaborated with the Service Enablement team to architect, develop, and document reusable Terraform modules for provisioning various AWS services, contributing to scalable infrastructure practices and adherence to organizational standards.

Syngenta — June 2023 – July 2024
Cloud Engineer

- Migrated **100+ AWS accounts** to Terraform Cloud, replacing manual provisioning with version-controlled IaC that eliminated configuration drift and gave every team a consistent, auditable, Git-reviewed deployment path.
- Cut **$500K** in annual AWS spend by rightsizing over-provisioned **EC2 and RDS instances**, scheduling non-production environments to power down outside business hours, and decommissioning idle volumes, snapshots, and load balancers.
- Built reusable **Terraform modules** and CI/CD deployment pipelines that cut deployment time 40% and sustained 99.9% uptime for 10,000+ users, letting application teams ship low-latency services without provisioning infrastructure by hand.
- Implemented **Sentinel policy-as-code** across AWS and GCP in Terraform Cloud, automating compliance and security checks at plan time to block non-compliant resources before deployment and reducing manual review effort by 70%.
- Migrated **DynamoDB** tables across AWS accounts using on-demand backup and restore, validating record counts and throughput configuration after cutover to complete the transition with zero data loss and minimal downtime.

Syngenta — Aug 2022 – Jun 2023
Software Engineering Intern

- Provisioned cloud infrastructure for 20+ applications using **AWS** services such as **EC2**, **S3, RDS, DynamoDB, API Gateway, Lambda, ECS, ECR, and ELB** ensuring efficient deployment and scalability.
- Planned and implemented Terraform modules for AWS services that standardized infrastructure management across departments, now utilized by over 50 teams, providing a reliable framework for secure and efficient cloud resource management.
- Developed and documented a solution for AWS database services (DynamoDB, Redshift, DocumentDB, Timestream), which has been successfully adopted by teams, improving performance by 30% and enhancing data processing efficiency by 25%.

PROJECT EXPERIENCE

Dynamic Weight Learning for Self-RAG, ***Team of 2*** — Aug 2025 – Dec 2025

- Built a dynamic weighting framework for Self-RAG pipelines that re-ranks and adaptively weights retrieved passages by query relevance, replacing static weighting that treated all retrieved documents equally regardless of context.
- Implemented adaptive scoring in Python and PyTorch across **1,000+ query** evaluations, improving response accuracy and relevance by **~18%** while keeping inference overhead low through optimized retrieval scoring logic.

Omnireasoner, Knowledge Graph Reasoning System, ***Team of 4*** — Aug 2025 – Mar 2026

- Built a knowledge-graph-augmented reasoning pipeline combining graph traversal with **LLM** prompting over a **10K+ entity graph**, improving factual consistency on complex multi-hop question answering.
- Generated interpretable reasoning paths through entity linking and graph traversal, reducing hallucinated responses by **~15% in evaluation** by grounding answers in retrieved graph relations rather than model priors.